Section 3.5
Nonhomogeneous Equations and Undetermined Coefficients

We have spent a lot of time studying homogeneous linear equations but now we will turn out attention to Non-Homogeneous equations. That is linear differential equations of the form:

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_2 y'' + a_1 y' + a_0 y = f(x)$$

We saw in section 3.2 that the full solution to the non-homogeneous equation was actually a sum of two solutions.

$$y = y_c + y_p$$

Where y_c is called the complementery solution and solves the homogeneous problem:

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_2 y'' + a_1 y' + a_0 y = 0$$

While y_p is called the particular solution and solves directly the nonhomogeneous proboblem

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_2 y'' + a_1 y' + a_0 y = f(x)$$

To see why we need the complementary solution recall that linear in y gives us

$$G(y_1 + y_2) = G(y_1) + G(y_2)$$

Here $G(y) = a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_2 y'' + a_1 y' + a_0 y$

So then if y_c solves the homogeneous problem $G(y) = 0$ and y_p solves the nonhomogeneous problem $G(y) = f(x)$ then $Y = y_c + y_p$ into $G(y)$ would give….

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In section 3.3 we studied the methods to solve the complementary solution y_c

DE: $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_2 y'' + a_1 y' + a_0 y = 0$

Characteristic Equation: $a_n r^n + a_{n-1} r^{n-1} + \cdots + a_2 r^2 + a_1 r + a_0 = 0$

Real distinct roots give parts of the general solution of the form,

$y_i = c_i e^{r_i x}$

Real repeated roots of multiplicity k give parts of the general solution of the form,

$y_i = c_i e^{\rho x}, y_{i+1} = c_{i+1} x e^{\rho x}, y_{i+1} = c_{i+2} x^2 e^{\rho x}, \dots, y_{i+(k-1)} = c_{i+(k-1)} x^{k-1} e^{\rho x}$

Complex roots $a + bi, a - bi$ with will come in complex conjugate pairs give parts of the general solution of the form

$y_i = e^{ax}\big(c_i \cos(bx) + d_i \sin(bx)\big)$

Complex repeated roots with multiplicity k will give parts of the general solution of the form

$y_i = e^{ax}\big(c_i \cos(bx) + d_i \sin(bx)\big), y_{i+1} = x e^{ax}\big(c_{i+1} \cos(bx) + d_{i+1} \sin(bx)\big),$
$y_{i+2} = x^2 e^{ax}\big(c_{i+2} \cos(bx) + d_{i+2} \sin(bx)\big), \dots, y_{i+(k-1)} = x^{k-1} e^{ax}\left(c_{i+(k-1)} \cos(bx) + d_{i+(k-1)} \sin(bx)\right)$

The full solution will be a linear combination of these pieces!

Now we will learn how to find y_p for the non-homogeneous equataions. The method we will study is typically called the **Method of Undetermined Coefficients.**

For a non-homogeneous equation $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_2 y'' + a_1 y' + a_0 y = f(x)$

The general strategy will be to look at the form of $f(x)$ and guess a linear combination of functions $(y_p = c_1 g_1(x) + c_2 g_{2(x)} + \cdots)$ for which when derivatives are taken, multiplied by the coefficients in the DE, and added up might produce $f(x)$. We will find the coefficents of the linear combination of guess functions so that our guess indeed does produce $f(x)$ when it is plugged into the differential equaiton.

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Let's start with an example. Find the solution to the differential equation

$$y'' + 2y' + y = 2x^2 + 1$$

What function might make a good guess for y_p?

Let's plug this into the differential equation and find the coefficients.

Also we can't forget to solve the complementary equation: $y'' + 2y' + y = 0$

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Nonhomogeneous Equations and Undetermined Coefficients

For nonhomogeneous equations

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_2 y'' + a_1 y' + a_0 y = f(x)$$

Where $f(x) = A_n x^n + A_{n-1} x^{n-1} + \cdots + A_0$ where some of the coefficients may be zero the guess you want to make for your y_p is

$$y_p = C_n x^n + C_{n-1} x^{n-1} + \cdots + C_0$$

(Provided none of the terms are duplicates of any part of the complementary solution)

You then find the $C's$ by plugging y_p in to the differential equation equating both sides and matching coefficients.

Let's look at another example. Find the particular solution to the differential equation

$$3y'' + y' - 2y = 2\cos(x)$$

What might be a good guess for y_p?

Let's find the undetermined coefficients.

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Nonhomogeneous Equations and Undetermined Coefficients

In general for nonhomogeneous equations

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_2 y'' + a_1 y' + a_0 y = f(x)$$

Where $f(x) = A_1 \cos(x) + B_1 \sin(x) + A_2 \cos(2x) + B_2 \sin(2x) + \cdots$ where some of the coefficients may be zero the guess you want to make for your y_p is

$$y_p = C_1 \cos(x) + C_2 \sin(x) + C_3 \cos(2x) + C_4 \sin(2x) + \cdots$$

(Provided none of the terms are duplicates of any part of the complementary solution)

You then find the $C's$ by plugging y_p in to the differential equation equating both sides and matching coefficients.

Another example: Find the particular solution to the differential equation:

$$y'' + 16y = e^{3x}$$

What might be a good choice for y_p?

Let's find the undetermined coefficients.

Section 3.5
Nonhomogeneous Equations and Undetermined Coefficients

In general for nonhomogeneous equations

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_2 y'' + a_1 y' + a_0 y = f(x)$$

Where $f(x) = A_1 e^{r_1 x} + A_2 e^{r_2 x} + \cdots$ where some of the coefficients may be zero the guess you want to make for your y_p is

$$y_p = C_1 e^{r_1 x} + C_2 e^{r_2 x} + \cdots$$

(Provided none of the terms are duplicates of any part of the complementary solution)

You then find the $C's$ by plugging y_p in to the differential equation equating both sides and matching coefficients.

$$y'' - 4y = 2e^{2x}$$

Based on the above what might we try?

What happens when we do?

Let's check the complementary solution, the one that solves $y'' - 4y = 0$

Any guess as to what we might try to do to fix this?

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Nonhomogeneous Equations and Undetermined Coefficients

$$y'' - 4y = 2e^{2x}$$

Let's try $y_p = Axe^{2x}$

So what do we do when parts of our guesses for our particular solution end up also satisfying the homogeneous equation?

Consider the differential equation: $(D-2)^3 y = e^{2x}$, what is the complementary solution?

Recall that $(D-r)^k u(x)e^{rx} = e^{rx}[D^k u(x)]$ before we chose polynomial terms of order $k-1$ for $u(x)$ so that k derivatives of them would give us zero, which we want for the homogeneous solution. However now we want something that won't go away for our parlticular solution. In this case we want something that will just go to a constant. If we have k derivatives perhaps a power of x that will reduct to a constant after that. Like x^k

Since if $u(x) = x^k$ then $(D-r)^k u(x)e^{rx} = e^{rx}[D^k u(x)]$ $= e^{rx}\left[D^k x^k\right] = e^{rk}$

So if our complementary solution is $y_c = c_1 e^{2x} + c_2 xe^{2x} + c_3 x^2 e^{2x}$ and our particular solution also needs a e^{2x} what might we try?

What happens when we do?

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So what we do when we have a term in our guess that is in the complementary solution is multiply it by the lowest power of x that we can so that it is nolonger a repeated term.

Suppose $y_c = c_3 cos(x) + c_4 \sin(x) + c_4 \cos(4x) + c_5 \sin(4x) + c_6 x cos(4x) + c_7 x sin(4x)$ is the complementary solution to the differential equation:

$$G(y) = 4\cos(x) + 2\sin(2x)$$

What guess would we make for our particular solution?

So let's summarize the types of guesses we might make for different $f(x)'s$

$f(x)$	$y_p(x)$
$P_m(x) = b_0 + b_1 x + b_2 x^2 + \cdots + b_n x^n$	$x^s(A_0 + A_1 x + A_2 x^2 + \cdots + A_n x^n)$
$\text{acos}(kx) + b\text{sin}(kx)$	$x^s(Acos(kx) + Bsin(kx))$
$e^{rx}(\text{acos}(kx) + bsin(kx))$	$x^s e^{rx}(\text{acos}(kx) + bsin(kx))$
$(b_0 + b_1 x + b_2 x^2 + \cdots + b_n x^n)e^{rx}$	$x^s(A_0 + A_1 x + A_2 x^2 + \cdots + A_n x^n)e^{rx}$
$(b_0 + b_1 x + b_2 x^2 + \cdots + b_n x^n)(\text{acos}(kx) + bsin(kx))$	$x^s[(A_0 + A_1 x + A_2 x^2 + \cdots + A_n x^n)\cos(kx) + (B_0 + B_1 x + B_2 x^2 + \cdots + B_n x^n)\sin(kx)]$

Where *s* is the smallest integer such that no term in the particular solution is an exact repeat of any term in the complementary solution.

What do we do when multiple of these types of terms show up in a sum on the other? Well we sum our guesses.

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Nonhomogeneous Equations and Undetermined Coefficients

Find the general form of the particular solution of : $y^{(3)} + 9y' = \cos(3x) + xsin(x) + x^2e^{2x}$

Let's make use of our table…..

So let's summarize the types of guesses we might make for different $f(x)'s$

$f(x)$	$y_p(x)$
$P_m(x) = b_0 + b_1x + b_2x^2 + \cdots + b_nx^n$	$x^s(A_0 + A_1x + A_2x^2 + \cdots + A_nx^n)$
$\text{acos}(kx) + b\text{sin}(kx)$	$x^s(Acos(kx) + Bsin(kx))$
$e^{rx}\big(\text{acos}(kx) + bsin(kx)\big)$	$x^s e^{rx}\big(\text{acos}(kx) + bsin(kx)\big)$
$\big(b_0 + b_1x + b_2x^2 + \cdots + b_nx^n\big)e^{rx}$	$x^s\big(A_0 + A_1x + A_2x^2 + \cdots + A_nx^n\big)e^{rx}$
$\big(b_0 + b_1x + b_2x^2 + \cdots + b_nx^n\big)\big(\text{acos}(kx) + bsin(kx)\big)$	$x^s\big[\big(A_0 + A_1x + A_2x^2 + \cdots + A_nx^n\big)\cos(kx)$ $+ \big(B_0 + B_1x + B_2x^2 + \cdots + B_nx^n\big)\sin(kx)\big]$

Where *s* is the smallest integer such that no term in the particular solution is an exact repeat of any term in the complementary solution.

First find that complementary solution to check for possible repeats…

Section 3.5
Nonhomogeneous Equations and Undetermined Coefficients

Determine the appropriate form of the particular solution for the differential equation

$$(D-2)^3(D^2+9)y = x^2e^{2x} + xsin(3x)$$

So let's summarize the types of guesses we might make for different $f(x)'s$

$f(x)$	$y_p(x)$
$P_m(x) = b_0 + b_1x + b_2x^2 + \cdots + b_nx^n$	$x^s(A_0 + A_1x + A_2x^2 + \cdots + A_nx^n)$
$\text{acos}(kx) + b\text{sin}(kx)$	$x^s(Acos(kx) + Bsin(kx))$
$e^{rx}(\text{acos}(kx) + bsin(kx))$	$x^se^{rx}(\text{acos}(kx) + bsin(kx))$
$(b_0 + b_1x + b_2x^2 + \cdots + b_nx^n)e^{rx}$	$x^s(A_0 + A_1x + A_2x^2 + \cdots + A_nx^n)e^{rx}$
$(b_0 + b_1x + b_2x^2 + \cdots + b_nx^n)(\text{acos}(kx) + bsin(kx))$	$x^s[(A_0 + A_1x + A_2x^2 + \cdots + A_nx^n)\cos(kx) + (B_0 + B_1x + B_2x^2 + \cdots + B_nx^n)\sin(kx)]$

Where *s* is the smallest integer such that no term in the particular solution is an exact repeat of any term in the complementary solution.

Complementary solution check……

Section 3.5
Nonhomogeneous Equations and Undetermined Coefficients

Example: Solve the initial value problem

$$y'' - 3y' + 2y = 3e^{-x} - 10\cos(3x)\,,\, y(0) = 1, y'(0) = 2$$

Section 3.5
Nonhomogeneous Equations and Undetermined Coefficients

Another method to finding particular solutions to differential equations is that of variation of parameters.

Let's consider the differential equation $G(y) = y'' + p(x)y' + q(x)y = f(x)$ and suppose it has complementary solution $y_c = c_1y_1 + c_2y_2$. Armed with this let's derive a formula for the particular solution y_p.

We will begin by assuming that the particular solution can be written involving the complementary solution functions and has the form:

$$y_p = u_1(x)y_1 + u_2(x)y_2$$

If we take the needed derivatives to plug into our differential equation we get.

$y_p' =$

$y_p'' =$

One requirement we have on y_p is that $G(y) = f(x)$ but here we have two unknown functions $u_1 and\ u_2$ so we need more requirements to find them. Having nothing else we are free to seek the simplest solution we can. It would be nice if in trying to find our unknown funcrtions we did not have to think about second derivatives so let us impose the requirement that

$$u_1'y_1 + u_2'y_2 = 0$$

Then our derivatives become

$y_p' =$

$y_p'' =$

Section 3.5
Nonhomogeneous Equations and Undetermined Coefficients

Considering differential operator $G(y) = y'' + p(x)y' + q(x)y$ let's plug in $y_p = u_1(x)y_1 + u_2(x)y_2$

$$G(y) = y_p'' + p(x)y_p' + q(x)y_p$$

$$G(y_p) = u_1y_1'' + u_2y_2'' + u_1'y_1' + u_2'y_2' + p(x)(u_1y_1' + u_2y_2') + q(x)(u_1y_1 + u_2y_2)$$

Let's collect all of the terms with no derivatives of u

$$G(y_p) = (u_1y_1'' + u_2y_2'' + p(x)(u_1y_1' + u_2y_2') + q(x)(u_1y_1 + u_2y_2)) + u_1'y_1' + u_2y_2'$$

$$G(y_p) = \left(u_1y_1'' + p(x)(u_1y_1') + q(x)(u_1y_1)\right) + \left(u_2y_2'' + p(x)(u_2y_2') + q(x)(u_2y_2)\right) + u_1'y_1' + u_2y_2'$$

$$G(y_p) = \left(u_1\left(y_1'' + p(x)(y_1') + q(x)(y_1)\right)\right) + \left(u_2\left(y_2'' + p(x)(y_2') + q(x)(y_2)\right)\right) + u_1'y_1' + u_2'y_2'$$

Now recall that y_1, y_2 solve the homogeneous problem $y'' + p(x)y' + q(x)y = 0$ so $G(y_p)$ is

$$G(y_p) = u_1'y_1' + u_2'y_2'$$

What we want of our particular solution is for $G(y_p) = f(x)$ so we can say we require that

$$u_1'y_1' + u_2'y_2' = f(x)$$

We can combine this with the previous requirement that

$$u_1'y_1 + u_2'y_2 = 0$$

To get two equations and two unknowns (ther derivatives of u_1, u_2). Then if we solve for those derivatives we simply integrate them to get u_1, u_2 and then we build $y_p = u_1y_1 + u_2y_2$

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Nonhomogeneous Equations and Undetermined Coefficients

$$u_1'y_1 + u_2'y_2 = 0$$
$$u_1'y_1' + u_2'y_2' = f(x)$$

Let's solve the system.

Written in matrix form this system is $\begin{bmatrix} y_1 & y_2 \\ y_1' & y_2' \end{bmatrix}\begin{bmatrix} u_1' \\ u_2' \end{bmatrix} = \begin{bmatrix} 0 \\ f(x) \end{bmatrix}$

Recall that if $Ax = b$ then $x = A^{-1}b$ and for a two by two matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $A^{-1} = \frac{1}{\det(A)}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

So assuming we have a solution for all x we have

$$\begin{bmatrix} u_1' \\ u_2' \end{bmatrix} = \frac{1}{y_1y_2' - y_1'y_2}\begin{bmatrix} y_2' & -y_2 \\ y_1' & y_1 \end{bmatrix}\begin{bmatrix} 0 \\ f(x) \end{bmatrix} = \frac{1}{W(y_1, y_2)}\begin{bmatrix} -y_2f(x) \\ y_1f(x) \end{bmatrix}$$

Where $W(y_1, y_2)$ is the wronskian of y_1, y_2

Therefore

$$u_1' = -\frac{y_2f(x)}{W(y_1, y_2)} \qquad u_2' = \frac{y_1f(x)}{W(y_1, y_2)}$$

So that Type equation here.

$$u_1 = -\int \frac{y_2f(x)}{W(y_1, y_2)}dx \qquad u_2 = \int \frac{y_1f(x)}{W(y_1, y_2)}dx$$

Thus the particular solution $y_p = y_1u_1 + y_2u_2$ becomes

$$y_p = -y_1\int \frac{y_2f(x)}{W(y_1, y_2)}dx + y_2\int \frac{y_1f(x)}{W(y_1, y_2)}dx$$

Section 3.5
Nonhomogeneous Equations and Undetermined Coefficients

If the non-homogeneous equation $y'' + p(x)y' + q(x)y = f(x)$ has complementary solution $y_c = c_1y_1 + c_2y_2$, then the particular soliution is given by

$$y_p = -y_1 \int \frac{y_2 f(x)}{W(y_1, y_2)} dx + y_2 \int \frac{y_1 f(x)}{W(y_1, y_2)} dx$$

Where $W(y_1, y_2)$ is the wronskian of the two linearly independent functions y_1, y_2

Right here we see how important the complementary solution is and why we can't just ignore it. It in fact defines the particular solution.

We did this for second order equation for ease, but this certainly applies to higher order linear nonhomogeneous equations.

For something like

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_2 y'' + a_1 y' + a_0 y = f(x)$$

With complementary solution $y_c = c_1y_1 + c_2y_2 + \cdots + c_ny_n$

We would take the form of y_p *to be* $y_p = u_1y_1 + u_2y_2 + \cdots + u_ny_n$ and play the same game we did with our second order equation and you would obtain

$$y_p = y_1 \int \frac{W_1 f(x)}{W(y_1, \dots, y_n)} dx + y_2 \int \frac{W_2 f(x)}{W(y_1, \dots, y_n)} dx + \cdots + y_n \int \frac{W_n f(x)}{W(y_1, \dots, y_n)}$$

Where $W_1, W_2, \dots W_i, \dots, W_n$ are the determinants of the matrix we get when replacing the ith column of the Wronskian by $(0,0,0, \dots, 1)$. Sometimes for higher order equations, it is just easier to make our guesses!